

Analysis of Seasonal Disease Patterns

Using Government Health Data: An Epidemiological Surveillance Study in Maharashtra

CANDIDATE & PROGRAMME

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PRIMARY SURVEILLANCE SOURCE

**NCDC / IDSP Weekly Outbreak
Bulletins**

Government of India Health Portals

EMPIRICAL SURVEILLANCE SCOPE

Maharashtra (36 Districts)

2022–2025 Baseline + 2026 W01–W32

Investigating Outbreak Seasonality in Maharashtra

PRIMARY RESEARCH INQUIRY

“What seasonal patterns are evident in selected disease outbreak reports in Maharashtra, and how do these patterns vary across diseases and seasons?”

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TEMPORAL CONCENTRATION

1

3-Month Concentration Ratio (CR3)

Determines whether outbreak events cluster within specific calendar months or disperse uniformly across the surveillance year.

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INTER-ANNUAL STABILITY

2

Kendall's W Coefficient of Concordance

Tests whether monthly disease ranking repeats reliably across four completed years (2022–2025) or exhibits volatile, random shifts.

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EVENT / VOLUME DIVERGENCE

3

Sensitivity & Outlier Decomposition

Evaluates how isolated mass-exposure outbreaks distort aggregate case volumes without altering underlying event frequency distributions.

From Weekly Government Reports to Structured Data



ADDITIVE NORMALIZATION // RAW STRINGS PRESERVED

Source Disease Nomenclature Never Overwritten

- 116 unique raw strings (e.g. 'Dengue?', 'Dengue fever', 'Dengue Shock') mapped to canonical labels.
- Original source string preserved in every record under 'disease_raw' for total reproducibility.
- Zero arbitrary alterations: ambiguous labels retained with full transformation audit rules.

FORENSIC AUDITABILITY // PDF PROVENANCE RETAINED

Every Tabular Record Linked to Source Bulletin

- Direct hyperlinkage to specific NCDC PDF filename and publication week number.
- Table page number and reporting district cross-verified against official archives.
- Automated integrity checks flag missing weeks, non-standard tables, and reporting anomalies.

Four-Year Outbreak Surveillance Baseline (Maharashtra)

539

OUTBREAK RECORDS

Investigated Clusters (2022–2025)

District surveillance investigations across
Maharashtra

21,955

REPORTED CASES

Enumerated Outbreak Patients

Aggregated across official cluster reports (5
primary families)

266

REPORTED FATALITIES

Surveillance Outbreak Deaths

Case fatality ratio: 1.21% across primary cohort

DENGUE

204 records

3,144 cases • 93
deaths

Peak: Oct (Aug–Oct CR3
50.5%)

ACUTE DIARRHEAL DISEASE

153 records

9,050 cases • 93
deaths

Peak: Jun–Aug (CR3 43.1%)

MALARIA

70 records

2,133 cases • 51
deaths

Peak: May–Jul (CR3 51.4%)

FOOD POISONING

69 records

5,927 cases • 29
deaths

Bimodal: Jan–Feb & Apr–
May

CHIKUNGUNYA

43 records

701 cases • 0 deaths

Dispersed: May, Jun, Sep

NOTE: Reported cases represent patients in investigated outbreak clusters. These figures do not measure population disease incidence or community prevalence.

Five Diseases. Five Distinct Temporal Signatures.

Calendar-month concentration curves derived from 539 investigated outbreak events

OBSERVED TEMPORAL CONCENTRATIONS

1. DENGUE

Late-Monsoon / Post-Monsoon Peak

October peak (40 events). 82.8% of all events occur Jun–Oct. Aug–Oct CR3 = 50.5%.

2. ACUTE DIARRHEAL DISEASE

Early Monsoon Plateau + Oct Rise

Sustained elevation Jun–Aug (18-20 events/mo) with secondary Oct rise. CR3 = 43.1%.

3. MALARIA

Pre-Monsoon & Early Monsoon

Concentrates earlier in May–July (12–14 events/mo). May–Jul CR3 = 51.4%.

4. FOOD POISONING

Bimodal Non-Monsoon Windows

Two distinct peaks: Jan–Feb and Apr–May. Demonstrates no monsoon concentration.

5. CHIKUNGUNYA

Low-Count Dispersed Distribution

Low baseline frequency without a single dominant month (CR3 = 41.9%).

BOUNDARY: Calendar concentrations reflect surveillance reporting; no climate/rainfall causality is inferred.

EXHIBIT: 4-YEAR MONTHLY EVENT DISTRIBUTIONS

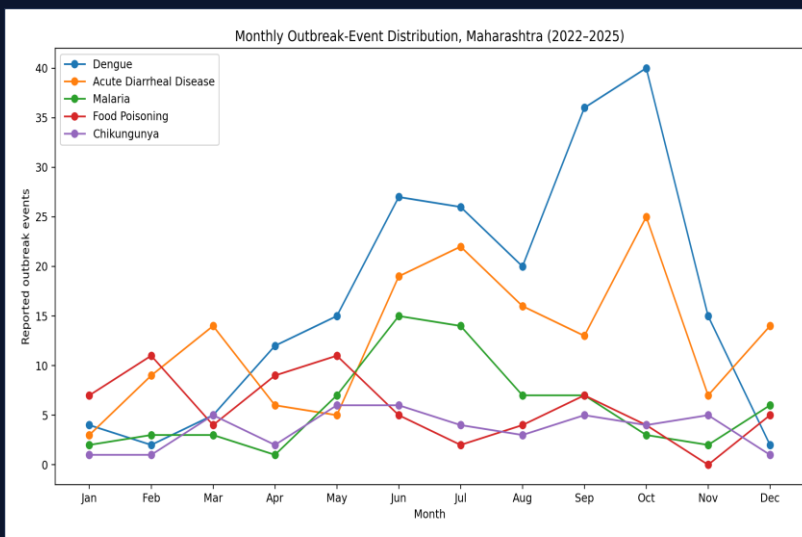


Figure: Monthly outbreak event counts aggregated across 2022–2025 (Phase 5B Analysis).

Observed Calendar Concentration & Kendall's W Stability

Measuring four-year monthly ranking consistency across 2022–2025 (tested at $\alpha = 0.05$)

KENDALL'S W CONCORDANCE ANALYSIS

DISEASE FAMILY	CR3 (TOP 3 MO)	W-STAT	p-VALUE
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Acute Diarrheal Dis.	43.1% (Jun–Aug)	0.530	0.016*
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Statistically significant inter-annual concordance ($p < 0.05$)

Dengue	50.5% (Aug–Oct)	0.405	0.086
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Non-significant (annual peak shifts between Sep and Oct)

Malaria	51.4% (May–Jul)	0.319	0.229
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Non-significant (early-monsoon elevation with variable timing)

Food Poisoning	44.9% (Apr/May/Feb)	0.260	0.380
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Non-significant (bimodal, sensitive to mass-case events)

Chikungunya	41.9% (May/Jun/Sep)	0.221	0.490
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Non-significant (low baseline frequency across the year)

* Kendall's W measures rank concordance (0 = no agreement, 1 = complete concordance). Tested at $\alpha = 0.05$ across 4 completed years.

EXHIBIT: INTER-ANNUAL STABILITY HEATMAP

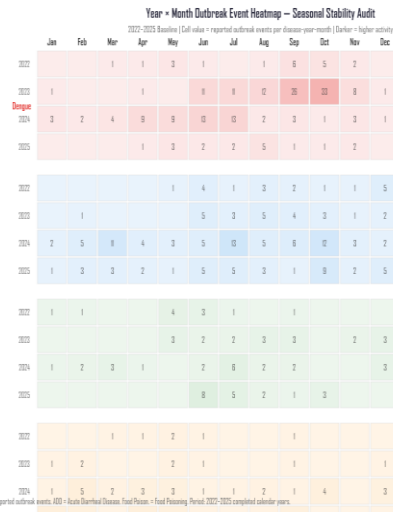


Figure: Kendall's W Rank Concordance across 2022–2025 (Phase 5C Analysis).

How Three Records Shape Four-Year Case Totals

Demonstrating the critical separation between outbreak frequency and patient enumeration

THREE MASS-EXPOSURE OUTBREAK RECORDS

651

KOLHAPUR DISTRICT

2024 Week 07

629

PARBHANI DISTRICT

2025 Week 06

335

SOLAPUR DISTRICT

2023 Week 06

COMBINED IMPACT: 1,615 CASES FROM 3 OUTBREAKS

74.7%

SHARE OF FEBRUARY CASES

1,615 of 2,162 Feb cases

27.3%

SHARE OF 4-YEAR TOTAL

1,615 of 5,927 total cases

15.9%

FEBRUARY EVENT RATIO

Only 11 of 69 events

CRITICAL METHODOLOGICAL TAKEAWAY:

Event frequency and patient case volumes measure fundamentally distinct phenomena. Without sensitivity testing, February would appear to be a massive seasonal transmission peak, when it is actually driven by 3 localized high-volume cluster reports. No causal assumptions (e.g. catering) are made.

EXHIBIT: SENSITIVITY DECOMPOSITION



Figure: Monthly Food Poisoning cases with vs. without 3 outlier events (Phase 5C Analysis).

2026 Surveillance Check (Weeks 01–32)

Evaluating out-of-sample data under explicit temporal truncation boundaries

PARTIAL-YEAR WINDOW // WEEKS 01–32 ONLY

45

RECORDS

W01–W32 primary cohort

1,118

CASES

Enumerated in clusters

0

DEATHS

Primary cohort to W32

1. DENGUE // TEMPORAL TRUNCATION BIAS

Dengue recorded 6 events (36 cases) up to Week 32. However, four-year baseline data demonstrates that 55.4% of all Dengue outbreaks occur after Week 32 (Sep–Dec). The 2026 observation window terminates before the historical surge window opens; week-32 totals are thus inherently incomplete.

2. FOOD POISONING // DIAGNOSTIC CLASSIFICATION EFFECT

Food Poisoning recorded 19 events (375 cases) in W01–W32, exceeding the historical early-year mean of 12.8 events (+49.0%). Notably, 100% of these 19 records were labeled 'Suspected' in source PDFs, indicating how syndromic reporting thresholds impact surveillance counts.

BOUNDARY: 2026 data is an out-of-sample check, not a validation or forecast.

EXHIBIT: 2026 SURVEILLANCE COMPARISON

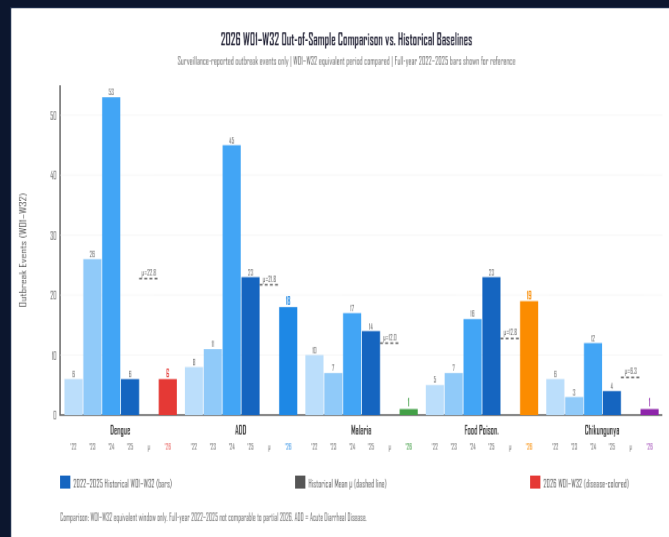


Figure: 2026 W01–W32 event counts compared to historical 2022–2025 baseline (Phase 5C Analysis).

Five Methodological Boundaries of This Analysis

Explicit scientific scope constraints ensuring academic rigor and defensibility

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SURVEILLANCE ≠ POPULATION INCIDENCE

Government outbreak bulletins capture officially investigated disease clusters through passive administrative channels. Mild, treated-at-home, or uninvestigated endemic cases across the broader community are not enumerated.

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NO CAUSAL OR CLIMATE ATTRIBUTION

This analysis identifies observed calendar-month outbreak concentrations only. Meteorological covariates (precipitation, humidity, temperature) were not co-modeled, precluding direct causal attribution to weather variables.

2

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DIAGNOSTIC & REPORTING HETEROGENEITY

Diagnostic thresholds vary across districts and seasons. Clinical syndromic labels (e.g. 'Suspected' food poisoning) coexist with serologically confirmed entries in source PDFs, producing inherent reporting variance.

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TEMPORAL TRUNCATION (2026 SURVEILLANCE)

The 2026 dataset spans Weeks 01–32 only. Diseases exhibiting second-half seasonal surge dynamics (notably Dengue, where 55.4% of historical events occur after W32) cannot be evaluated across this truncated window.

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INSTITUTIONAL SURVEILLANCE SCOPE

Surveillance records capture public-sector investigation alerts. Outpatient consultations and private-sector clinical encounters that do not trigger formal IDSP rapid-response team deployments are not captured in these data.

5

What the Four-Year Analysis Demonstrates

Summary of empirical findings across 539 investigated outbreak events in Maharashtra

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DISTINCT TEMPORAL SIGNATURES

The five disease families exhibit fundamentally different calendar distributions: Dengue concentrates in Jun–Oct (Oct peak); Malaria in May–Jul; ADD shows sustained Jun–Aug elevation with Oct secondary activity; Food Poisoning is bimodal (Jan–Feb, Apr–May); Chikungunya is dispersed.

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EVENT FREQUENCY VS. CASE VOLUME DIVERGENCE

Outbreak event counts and patient case totals can diverge substantially. A small number of mass-exposure records can dominate aggregate case volumes without altering underlying event frequency distributions.

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STATISTICALLY SIGNIFICANT INTER-ANNUAL CONCORDANCE

Acute Diarrheal Disease demonstrates the only statistically significant inter-annual monthly ranking stability (Kendall's $W = 0.530$, $p = 0.016$). All other disease families return non-significant concordance values.

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OUTLIER SENSITIVITY IN CASE TOTALS

Food Poisoning case totals are overwhelmingly driven by three mass-exposure records (1,615 cases) accounting for 74.7% of all February cases and 27.3% of four-year baseline cases across Maharashtra.

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PARTIAL-YEAR SURVEILLANCE BOUNDARIES

2026 Weeks 01–32 provides a partial out-of-sample comparison, not a full annual validation. Diseases with historical post-Week 32 surge patterns (notably Dengue) cannot be evaluated over this partial observation window.

SURVEILLANCE IMPLICATION

Historical concentration windows can assist public health authorities in prioritizing surveillance vigilance and logistical pre-positioning, while remaining strictly subject to reporting coverage constraints and source-data boundaries.